

CYBERSECURITY AWARENESS IN ONLINE EDUCATION A CASE STUDY ANALYSIS

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ABSTRACT

This study examines how well-versed Kyrgyz-Turkish Manas School students are in cybersecurity through online learning. An first-year, second-year, and PhD sample of 517 students from all of the university's faculties participated in the study. Despite the fact that there are a great deal of cyberattacks happening all over the world, our research reveals that the pupils had no knowledge of cybersecurity or the overall effects of assaults. By focusing on issues pertaining to harmful software, safe passwords, and social media security, an analysis of understanding of cybersecurity was conducted. It has been discovered that students' knowledge of cybersecurity is lacking, despite the fact that we live in a technologically advanced age where every aspect of our lives is connected to the net through distance learning. More study has led to the conclusion that security instruction should be provided to students in order to shield them from hacks and improve their use of the web.

Keywords: CyberSecurity, A Case Study, Online Education, Students.

I. INTRODUCTION

Cyber security has started to become extremely important in both individuals and states alike due to the growth of tech and the infiltration of cyberspace into every area of daily life. Although these advances have made our lives easier, the rise in cyberattacks has necessitated the adoption. The usage of laptops in attacks has not changed. Crimes are defined in this sense as crimes carried out on computers. in the US Bureau of Justice, a cybercrime is any violation of the law that requires computer technology of security measures. The forms of cyberattacks, or the harmful use of the internet age, have altered over the past 20 years, which is another key point. This has caused authors to use new "cyber" terms and risks. Hathaway and colleagues describe a cyberattack as "any step taken to undermine the operation of a computer networks for a political or national security purpose." The most fundamental inquiry to make is, "Does this term accurately define cyberattacks in the modern era?" Saying that hacks are exclusively conducted for political reasons is no longer sufficient when attempting to figure out the nature of cyberattacks. This is due to the evolution of new cyber concepts that have altered the character of cyberattacks knowledge for its commission, investigation, or prosecution. One the one hand, it's critical to define cyber security. The International Telecommunications Union (ITU) defines cyber security as a collection of tools, policies, security concepts, security protects, rules, risk-taking approaches, actions, training, best practises, assurance, and technologies that can be used to protect an organization's and user's assets in the cyberspace, despite the fact that there is no universal definition for the term. In order to protect an organization's and user's assets from relevant security dangers in the cyber environment, cyber security aims to achieve and maintain their security properties.

II. LITERATURE SURVEY

This study examines how well-versed Kyrgyz-Turkish Manas University students are in cybersecurity through online learning. An college, graduate, and PhD sample of 517 people from all of the university's faculties participated in the study. Despite the fact that there are a great deal of cyberattacks happening all over the world, our research reveals that the pupils had no interest in cybersecurity or the overall effects of cyberattacks. By focusing on issues pertaining to harmful software, safe passwords, and social media security, an analysis of understanding of cybersecurity was conducted. It has been discovered that students' knowledge of cybersecurity is lacking, despite the fact that we live in a technologically advanced age where every aspect of our lives is connected to the internet through distance learning. This study measures the level of cybersecurity

knowledge attained by Kyrgyz-Turkish Manas college pupils through distance learning. A total of 517 college, master's, and PhD students from all of the university's faculties made up the sample for the study. Our study's findings demonstrate that students' lack of awareness about cybersecurity and the overall effects of cyberattacks is despite the fact that there are numerous cyberattacks taking place all over the world. By focusing on hazardous software, secure passwords, and social media safety questions about cybersecurity knowledge were put together. As COVID-19 took hold and groups and classrooms all over the world had to close their doors, online learning started to become the standard. As of April 2020, 186 countries and more than 1.2 billion students were affected by educational institution closures. The school system there has also changed as a result of this pandemic. Online learning is becoming more popular both academics and students because to free and distance learning platforms. The cognitive computing and human intelligence revolution is known as "education 5.0." The main obstacle that one will encounter as the world moves in this direction is how to deal with the risks involved with safety online in the digital age. In order to protect against hacker attacks online, information security awareness might be crucial. The main objective of this paper is to analyse the level of information security awareness, associated risks, and overall effect on the institutions among professors, researchers, freshmen, and employees in schools in the Middle East. The findings show that those polled lack the necessary knowledge and comprehension of the significance of information security concepts and how they are used in their daily job. The Web is becoming a more integral part of many people's, organisations', and countries' daily lives. It has, in large part, positively impacted how people speak. Additionally, it has set up novel business possibilities and given countries the chance to conduct their governments online. Cyber comes with a lot of risks even if it provides an unending array of services and opportunities.

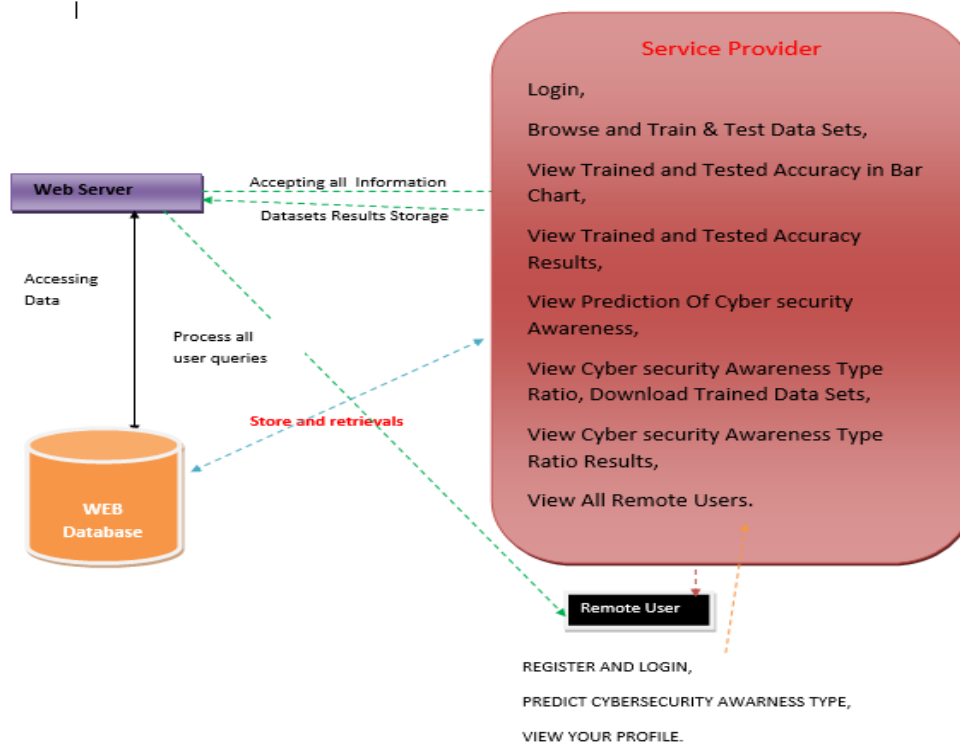


Fig.1 Proposed Architecture

III. EXISTING WORK

To gauge the degree of cybersecurity expertise among pupils and instructors, several research have been carried out. For instance, Ismailova and Muhametjanova's study of 172 participants' acquaintance with the risk of cybercrime in the Kyrgyz Republic. The findings indicate that the students had little knowledge of cybercrime. Another survey measuring cybersecurity knowledge among people aged 8 to 21 was conducted in New Zealand in 2016. Trimula, Sarrafzadeh, and Pang carried out this. The majority of pupils, according to the authors, were unaware of the existence of cyber dangers and were confused with the phrase cybersecurity.

Ahmed et al. investigated Bangladeshi citizens' knowledge of cybersecurity. According to their research, the sample lacked sufficient cybersecurity knowledge. The authors suggested that a manual be written so that people can consciously consider cyber. A survey from the Yobe University's Department of Computer Science revealed that while pupils were aware of security, they were unsure on how to safeguard their own data.

IV. PROPOSED METHODOLOGY

Data safety awareness, also known as awareness regarding cybersecurity, has grown in importance in the modern world. This topic, which has an impact on all facets of daily life, is the focus of a growing number of research. To properly comprehend the topic, it is crucial to begin by defining cybersecurity awareness. According to Shaw et al., the idea is "the level of users' understanding of the significance of information security and their duty and acts to exercise sufficient levels of info security control to protect the company's data and networks.". The definition makes it clear that there are two approaches to analyse the key issues. It first emphasises the significance of privacy and the associated obligations. Second, it aims to understand and employ safety control practises at a sufficient level to safeguard the data.

Privacy awareness, according to Hwang et al., is a phenomena that seeks to make consumers aware of potential security flaws or issues so they can react appropriately. Naturally, it also aims to keep users' minds constantly on the internet security phenomenon. comparable findings were made by Khan et al. to Hwang. According to Khan et al., information security awareness is the state in which users are aware of security issues and behave in line with established rules.

V. IMPLIMENTATION

Service Provider

The Service Provider must enter an adequate password and user name to log in to this module. After successfully logging in, he can perform a number of actions, including logging in, browsing, and training and testing data sets. View Trained and Tested Accuracy in Bar Chart, View Trained and Tested Accuracy Results, View prediction of Cybersecurity Awareness, View Cybersecurity Awareness Type Ratio, Download Trained Data Sets, View Cybersecurity Awareness Type Ratio Results, View all Remote Users.

View and Authorize Users

A The list of people who have filed can be seen by the owner in this module. The admin may examine the user's data within this, including user name, email address, and satisfy, and admin can also allow users.

Remote User

It has n numbers of users who use this module. When doing any tasks, the user should register. Once a user registers, the database will record their information. After fully signing up he must log in with an allowed user ID and password. After successfully logging in, the user can perform a number of actions, including REGISTER AND LOGIN, PREDICT CYBERSECURITY AWARENESS TYPE, and VIEW YOUR PROFILE.

VI. CONCLUSION

When the survey findings from students at Kyrgyz Turkish Manas University were analysed, it became clear that the majority of the participants lacked adequate awareness of internet use and cyberthreats. They were also discovered to lack technical expertise on a variety of topics, such as what the sites they visit have secure certificates and whether a hacker might get their information by tricking them. It would not be proper to deliver this knowledge exclusively in the parts that provide technical education, as cyber dangers affect those of all educational backgrounds. The findings of this study also demonstrate that those who received cyber security instruction had greater proficiency in computer usage and fundamental network security topics.

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